

Total no. of Pages: 02

Roll no.

Degree: B-Tech Semester: 3rd (Backlog)

SUMMER SEMESTER (END-SEMESTER) EXAMINATION, JULY 2024

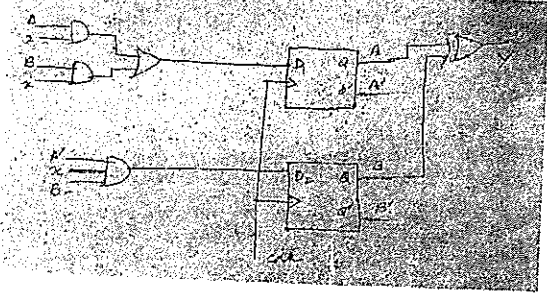
Course Code: ICECC08/EECC08

Course Title: Digital Circuit and System

Time: 03 Hours

Max. Marks: 50

Note: - Attempt all the five questions. Missing data/ information (if any), maybe suitably assumed & mentioned in the answer.

Q.No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	Design a four-bit binary to Gray code converter using the basic gates.	5	CO1
1b	Minimize the following function using K maps (1) $F(W,X,Y,Z) = \Sigma(1,3,4,5,6,7,9,12,13)$ (2) $F(W,X,Y,Z) = \pi((1,5,6,7,11,12,13,15))$	5	CO1
1c	Implement the following function using 8X1 multiplexer $F(A,B,C,D) = \Sigma(0,1,2,5,7,8,9,15)$	5	CO1
Q 2	Attempt any 2 parts of the following		
2a	Realize a JK flip flop using SR flip flop.	5	CO2
2b	Implement a 2 bit binary comparator comparing the bits $B=B_1B_0$ with $A=A_1A_0$	5	CO2
2c	Perform the following conversions (1) (10111) Gray to binary (2) (2376) octal to decimal (3) (ABC) hexadecimal to binary (4) (6678) decimal to excess 3 code	5	CO2
Q 3	Attempt any 2 parts of the following		
3a	Analyse the given circuit and derive it's state diagram and state table 	5	CO3

3b	Convert the given Moore machine into the Mealy machine	5	CO3																				
	<table border="1"> <thead> <tr> <th>PS</th><th>NS X=0</th><th>NS X=1</th><th>Output</th></tr> </thead> <tbody> <tr> <td>A</td><td>B</td><td>A</td><td>0</td></tr> <tr> <td>B</td><td>C</td><td>D</td><td>1</td></tr> <tr> <td>C</td><td>B</td><td>C</td><td>1</td></tr> <tr> <td>D</td><td>A</td><td>B</td><td>0</td></tr> </tbody> </table>	PS	NS X=0	NS X=1	Output	A	B	A	0	B	C	D	1	C	B	C	1	D	A	B	0		
PS	NS X=0	NS X=1	Output																				
A	B	A	0																				
B	C	D	1																				
C	B	C	1																				
D	A	B	0																				
3c	Write the VHDL code for a full adder circuit.	5	CO3																				
Q 4	Attempt any 2 parts of the following																						
4a	Explain with the help of a suitable diagram the working of a CMOS 2 input NAND and NOR gate.	5	CO4																				
4b	Differentiate between SRAM and DRAM.	5	CO4																				
4c	Write the VHDL code for a 4X1 multiplexer.	5	CO4																				
Q 5	Attempt any 2 parts of the following																						
5a	Subtract the following using 2's compliment method 1. (39)-(-21) 2. (19)-(+43)	5	CO5																				
5b	Write a short note on any 2 of the following (1) VHDL coding styles (2) Properties of logic families (3) SRAM and DRAM (4) CPLD and FPGA	5	CO5																				
5c	Explain with the help of block diagrams the working of the 4 different types of registers.	5	CO5																				